

MUNTASIR MAMUN

AI ENGINEER • GENERATIVE AI RESEARCHER

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PROFESSIONAL SUMMARY

AI Engineer and Undergraduate Researcher specializing in Machine Learning, Generative AI, Large Language Models (LLMs), and Computer Vision. Experienced in engineering end-to-end RAG pipelines, multi-agent LLM systems, diffusion architectures, and high-performance serverless GPU backends. Proven ability to bridge theoretical deep learning research with scalable production engineering.

TECHNICAL SKILLS

Artificial Intelligence: Machine Learning • Deep Learning • Computer Vision • Generative AI • Large Language Models (LLMs) • AI Agents • Retrieval-Augmented Generation (RAG) • Diffusion Models • Model Fine-Tuning

Frameworks & Libraries: PyTorch • TensorFlow • Scikit-learn • Hugging Face • LangChain • ComfyUI • OpenCV • NumPy • Pandas • FastAPI

Cloud & DevOps: Docker • Git • GitHub • AWS • Google Cloud Platform (GCP) • DigitalOcean • GPU Infrastructure • Model Deployment

Programming: Python • C++ • Java • SQL

RESEARCH EXPERIENCE

Undergraduate AI Researcher

BUP AI Research Lab

- **Deep Learning Signal Demodulation for Wireless Communication (Ongoing):** Investigating deep neural network architectures to perform robust signal demodulation under severe Rayleigh fading and noisy channel conditions. Benchmarking custom PyTorch model performance against conventional communication techniques using BER and SER metrics.
- **Diffusion Architecture Optimization for Virtual Try-On (Ongoing):** Experimenting with latent diffusion models and garment-alignment pipelines tailored for high-fidelity virtual try-on of traditional apparel. Evaluating identity-preservation mechanisms, pose-invariant attention maps, and optimizing reverse-sampling steps to reduce latency.

SELECTED PROJECTS

VESTON – AI Fashion Virtual Try-On Platform

Architected a full-stack Generative AI fashion e-commerce application utilizing advanced image-conditioned diffusion models.

- Engineered a virtual try-on pipeline integrating IDM-VTON and Hugging Face diffusers to execute realistic clothing transfers.
- Developed an asynchronous RESTful backend in FastAPI, optimizing request queuing and memory allocation for heavy generative workloads.
- Deployed models on serverless GPU infrastructure, reducing idle operational costs while maintaining low-latency inference.

Technologies: Python, PyTorch, Hugging Face, IDM-VTON, FastAPI, Docker, GPU Infrastructure

Metro Rail AI Assistant

Designed a multi-modal transit automation system combining speech recognition, route finding, and conversational intelligence.

- Constructed a voice-to-ticket booking pipeline integrating OpenAI Whisper for high-accuracy automatic speech recognition (ASR).
- Designed a LangChain-driven RAG architecture over vector storage to answer complex, real-time transit queries deterministically.
- Formulated intent-classification workflows to parse natural language queries into accurate topological route and fare calculations.

Technologies: Python, PyTorch, OpenAI Whisper, LangChain, RAG, Vector Databases, FastAPI

Multi-Agent AI Research Assistant

Built an autonomous AI agent framework designed to accelerate academic research workflows and literature analysis.

- Developed a multi-agent orchestration architecture using LangChain to conduct structured research, article retrieval, and cross-paper synthesis.
- Engineered prompt-chaining and context-compaction strategies to prevent context window overflow during multi-document processing.
- Implemented automated fact-verification and citation-tracking modules to eliminate LLM hallucinations in output summaries.

Technologies: Python, LangChain, LLMs, Vector Search, PyTorch, API Integration

RetailAI – Smart Shop Management System

Developed an automated retail inventory management tool driven by voice commands and structured language parsing.

- Implemented real-time voice-to-text processing using OpenAI Whisper to convert spoken inventory updates into structured DB entries.
- Reduced manual data entry friction through automated field parsing and structured schema insertion.

Technologies: Python, OpenAI Whisper, SQL, FastAPI, Docker

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor **AOSSIE & Global Repositories**

- Contributed pull requests (PRs) optimizing module performance, fixing edge-case bugs, and expanding core test suites.
- Conducted peer code reviews, adhering to strict CI/CD guidelines, standard Git workflows, and comprehensive unit testing.

EDUCATION & CERTIFICATIONS

Bangladesh University of Professionals (BUP)	B.Sc. in Information and Communication Engineering	Expected 2027
Saint Joseph Higher Secondary School	Higher Secondary Certificate (HSC)	Completed

Certifications: [AI Engineer for Data Scientists \(DataCamp\)](#) • [Data Analyst \(DataCamp\)](#) • Object-Oriented Programming in Java (DataCamp)